VitalStack

30-Day Java Interview Plan

A day-by-day plan for Java developer interviews and campus placements: core Java, OOP, SQL, Spring Boot and DSA, with a free lesson, practice and a mock test for every step.

30 days | ~2 hours a day | 4 mock-test checkpoints | 25 must-know questions

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How to use this plan

- **Two hours a day:** about 1 hour reading the lessons, 1 hour practising. Busy day? Do the practice, skip nothing else.
- **Every link opens the free lesson, problem or mock test** on vitalstack.co.in. Sign in so your progress is saved.
- **Tick the box** at the end of each day. Missed a day? Continue tomorrow; don't try to do two.
- **Checkpoints (days 7, 14, 21, 28):** take the mock test, then revise only the lessons it lists under “Revise these lessons”.
- **Say answers out loud.** Interviews test explaining, not only knowing. After each topic, explain it in one minute without notes.
- **Already know a week well?** Take its mock test first. Scored 18/20 or more? Move on.

By day 30 you will have three projects on GitHub (Student Grade Manager, Library System or your own OOP project, Course Enrollment API), around 20 classic problems solved, and the 25 questions on the last page answered in your own words.

Week 1 · Core Java

Get fast and accurate with the language itself.

DAY 1

Types, variables and operators

Read Java Variables and Data Types, Java Operators and Expressions

- Know every primitive type and its size; explain why `5 / 2` is `2` and `(int) 9.99` is `9`.
- Write a program that converts Celsius to Fahrenheit for 0–100 in steps of 10.

DAY 2

Strings

Read Java Strings

- Explain `==` vs `equals()`, why strings are immutable, and when to use `StringBuilder`.
- Reverse a string, count its vowels, and solve Valid Palindrome.

DAY 3

Conditions and loops

Read Java If, Else and Switch, Java Loops

- FizzBuzz, check if a number is prime, and sum the digits of a number.
- Rewrite one `if`/`else` chain as a `switch`.

DAY 4

Arrays and methods

Read Java Arrays, Java Methods

- Find the max, the second max and reverse an array in place.
- Be ready for: “Is Java pass-by-value or pass-by-reference?”

DAY 5

Collections: ArrayList and HashMap

Read Java ArrayList and Collections, Java HashMap and Maps

- Explain how a `HashMap` stores entries (buckets, hashing, collisions).
- Solve Two Sum and Contains Duplicate with a map or set.

DAY 6

Exceptions, lambdas and streams

Read Java Exception Handling, Java Lambdas and Streams

- Checked vs unchecked exceptions, and what `finally` guarantees.
- Use a stream to filter, map and collect a list of students by marks.

DAY 7

Checkpoint

Read Java Project: Student Grade Manager

- Take the Java MCQ Test. Target: 16/20. Revise every lesson it lists.
- Build the Student Grade Manager project and push it to GitHub.

Week 2 · OOP and SQL

The two topics asked in almost every Java interview.

DAY 8

Classes, objects, constructors, static

Read Java Classes and Objects, Java Static Members and Constructor Overloading

- Explain constructor overloading and what `static` belongs to.
- Model a `BankAccount` class with deposit and withdraw.

DAY 9

Encapsulation and inheritance

Read Java Encapsulation, Java Inheritance

- Why private fields with getters and setters? What does `super` do?
- Extend `BankAccount` into `SavingsAccount` with interest.

DAY 10

Polymorphism and abstraction

Read Java Polymorphism, Java Abstraction and Abstract Classes

- Overloading vs overriding, with your own example.
- Explain runtime (dynamic) method dispatch; watch the lesson's animation.

DAY 11

Interfaces, equals/hashCode, records

Read Java Interfaces, Java equals, hashCode and toString, Java Records, Enums and Sealed Classes

- Abstract class vs interface (including default methods).
- Why must `equals()` and `hashCode()` be overridden together?

DAY 12

SOLID and design patterns

Read SOLID Principles and Design Patterns in Java

- Explain each SOLID letter with one example.
- Write Singleton, Builder, Factory and Strategy from memory.

DAY 13

SQL queries

Read SQL SELECT, SQL WHERE, SQL ORDER BY, SQL Functions (COUNT, SUM, AVG), SQL GROUP BY

- `WHERE` vs `HAVING`; what `COUNT(*)` vs `COUNT(col)` returns with NULLs.
- Write the 2nd highest salary query and the count of employees per department.

DAY 14

JOINs, keys, indexes + checkpoint

Read SQL JOINs (INNER JOIN), SQL LEFT JOIN, SQL Constraints and Keys, SQL Indexes

- `INNER JOIN` vs `LEFT JOIN`; primary vs foreign key; when an index helps.
- Take the Java OOP and SQL mock tests. Target: 16/20 each.

Week 3 · Spring and Spring Boot

What most Java backend jobs actually use.

DAY 15

Spring core: IoC and dependency injection

Read Spring IoC Container and Dependency Injection, Spring @Configuration, @Bean, @Primary and @Qualifier, Spring Bean Scopes and Lifecycle

- What the IoC container does; why constructor injection is preferred over `@Autowired` on fields.
- Singleton vs prototype scope.

DAY 16

Your first REST API

Read Spring Boot Introduction, Build Your First REST API with Spring Boot

- What auto-configuration does; `@RestController` vs `@Controller`.
- Build a CRUD API for books with GET, POST, PUT and DELETE.

DAY 17

Spring Data JPA and JPA basics

Read Spring Data JPA - Repositories Without Code, EntityManager and Entity States, JPA Relationships - OneToMany, ManyToOne, ManyToMany

- Entity states (new, managed, detached, removed); lazy vs eager fetching.
- Connect the books API to a database with a repository.

DAY 18

Validation, errors and transactions

Read DTOs, Validation and Error Handling in Spring Boot, Spring @Transactional - Transactions Made Simple

- Why use DTOs; how `@Valid` and a global exception handler work.
- What `@Transactional` does, and when it silently doesn't.

DAY 19

Configuration and security

Read Spring Boot Configuration, Profiles and Logging, Spring Security with JWT in Spring Boot

- Profiles and `application.properties`; how JWT authentication works step by step.
- Secure the books API so only signed-in users can add books.

DAY 20

Testing and performance

Read Testing Spring Boot Applications, Relationships, Pagination and Performance in a Spring Boot API

- Unit test vs `@SpringBootTest`; what the N+1 query problem is and how to fix it.
- Add pagination to the list endpoint and one test per endpoint.

DAY 21

Checkpoint

Read Spring Boot Project: Course Enrollment API

- Take the Spring Boot MCQ Test. Target: 15/20.
- Build the Course Enrollment API project: your strongest resume project.

Week 4 · DSA and the interview

Patterns, not memorising. Then practise like it's real.

DAY 22

Big O, arrays and hashing

Read Big O Notation, Hashing: Hash Maps and Sets

- Solve Valid Anagram, Group Anagrams and Top K Frequent Elements.

DAY 23

Two pointers and sliding window

Read Two Pointers, Sliding Window

- Solve 3Sum, Container With Most Water and Longest Substring Without Repeating Characters.

DAY 24

Binary search and sorting

Read Binary Search, Merge Sort and Quick Sort

- Solve Search in Rotated Sorted Array and Koko Eating Bananas.

DAY 25

Stacks and linked lists

Read Stacks, Linked Lists

- Solve Valid Parentheses, Reverse Linked List and LRU Cache.

DAY 26

Trees and heaps

Read Binary Trees and Traversals, Heaps and Priority Queues

- Solve Binary Tree Level Order Traversal, Validate Binary Search Tree and Kth Largest Element in an Array.

DAY 27

Graphs

Read Graphs: BFS and DFS, Topological Sort

- Solve Number of Islands and Course Schedule.

DAY 28

Dynamic programming

Read Dynamic Programming (1D)

- Solve Climbing Stairs, House Robber and Coin Change.
- Take the DSA MCQ Test.

DAY 29

Mock interview day

Read The DSA Interview Playbook

- 45 minutes, one medium problem, explain your approach out loud before coding.
- Take the Coding Interview MCQ Test. Practise “Tell me about yourself” and explain both projects in 2 minutes each.

DAY
30

Final revision

Read Your notes and the lessons your tests flagged

- Retake the Java, Java OOP and Spring Boot tests; revise only what you got wrong.
- Skim the **Java** and **Spring Boot** cheat sheets. Then rest: sleep beats one more topic.

25 questions you should answer without notes

Asked again and again in Java interviews. Tick each one when you can explain it in under a minute. The link is the lesson that answers it.

- [ ] What is the difference between `==` and `equals()`?
  Java Strings
- [ ] Why are strings immutable, and when do you use `StringBuilder`?
  Java Strings
- [ ] JDK vs JRE vs JVM?
  Java Introduction
- [ ] How does a `HashMap` work internally? What happens on a collision?
  Java HashMap and Maps
- [ ] Checked vs unchecked exceptions? When does `finally` run?
  Java Exception Handling
- [ ] What are streams and lambdas? `map` vs `filter`?
  Java Lambdas and Streams
- [ ] What does `static` mean on a field and on a method?
  Java Static Members and Constructor Overloading
- [ ] What is encapsulation and why does it matter?
  Java Encapsulation
- [ ] Overloading vs overriding?
  Java Polymorphism
- [ ] Abstract class vs interface?
  Java Interfaces
- [ ] Why override `equals()` and `hashCode()` together?
  Java equals, hashCode and toString
- [ ] Explain SOLID. Implement a Singleton.
  SOLID Principles and Design Patterns in Java
- [ ] `WHERE` vs `HAVING`?
  SQL GROUP BY
- [ ] `INNER JOIN` vs `LEFT JOIN`?
  SQL LEFT JOIN
- [ ] Primary key vs foreign key? What is an index?
  SQL Constraints and Keys
- [ ] What is dependency injection? Why constructor injection?
  Spring IoC Container and Dependency Injection
- [ ] Bean scopes: singleton vs prototype?
  Spring Bean Scopes and Lifecycle
- [ ] What does `@Transactional` do?
  Spring @Transactional - Transactions Made Simple

- [ ] What does Spring Boot auto-configuration do?
  Spring Boot Introduction
- [ ] `@RestController` vs `@Controller`?
  Build Your First REST API with Spring Boot
- [ ] How do you validate input and return clean error messages?
  DTOs, Validation and Error Handling in Spring Boot
- [ ] How does JWT authentication work?
  Spring Security with JWT in Spring Boot
- [ ] Lazy vs eager loading? What is the N+1 problem?
  Relationships, Pagination and Performance in a Spring Boot API
- [ ] What is Big O? Time complexity of HashMap get and of binary search?
  Big O Notation
- [ ] How do you approach a problem you've never seen?
  The DSA Interview Playbook

Good luck! New lessons are added every week on **vitalstack.co.in**. Follow us for daily coding puzzles: Instagram and Threads **@vitalstack.ai**, X **@Vital_Stack**.